HiReNet: Hierarchical-Relation Network for Few-Shot Remote Sensing Image Scene Classification

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***Abstract*— Few-shot scene classification aims to develop models that can quickly adapt to new scenes with only a few labeled samples that are not present in training sets. In recent years, convolutional neural networks (CNNs) have made significant advancements in few-shot remote sensing image scene classification tasks. However, most existing approaches focus solely on utilizing high-level embeddings of remote sensing images to learn similarity relations, while neglecting intrinsic hierarchical representations that could be crucial in distinguishing scenes with substantial interclass similarities. To address this limitation, we propose a novel few-shot scene classification method for remote sensing images called hierarchical-relation network (HiReNet). This approach leverages the hierarchical features of a query sample and its corresponding support sample to learn discriminative representations. HiReNet consists of an embedding network and a relation network. The embedding network employs a Siamese architecture to extract representations, while the relation network utilizes these representations for classification. Within the relation network, we introduce a hierarchical relation learning (HRL) structure to capture the hierarchical relations among query and support samples. Additionally, to extract stronger features, we introduce a feature aggregation module that concatenates multilevel features and employs channel attention to re-weight these features. Experimental results demonstrate the superior performance of our HiReNet compared to several state-of-the-art few-shot scene classification methods.**

***Index Terms*— Convolutional neural networks (CNNs), few-shot learning, hierarchical relation, remote sensing images.**

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I. Introduction

REMOTE sensing scene classification aims to assign scene labels to remote sensing images and has played an important role in a series of tasks, such as land use and land cover determination, urban planning, geographical image retrieval, environment monitoring, and vegetation mapping [1].

Recently, deep learning-based methods have been widely applied in the field of remote sensing including object detection [2], super-resolution [3], [4], [5], and scene classification [6], [7], [8]. However, these methods often require massive labeled samples and the generalized performances would degrade heavily once training data are insufficient, especially in some novel and unknown situations.

To alleviate this problem, few-shot learning has been introduced in the field of remote sensing recently [9], [10], [11], [12], [13]. Different from the usual framework, remote sensing scene classification with few-shot learning aims to learn a representative model on a large number of labeled samples and adapt to distinguish unseen classes with a very small number of examples [11], [14], [15], [16]. Specifically, Zhai et al. [11] proposed a few-shot remote sensing classification model based on meta-learning methods, to transfer the knowledge learning from a known dataset to a novel one. Li et al. [14] introduced a new few-shot classification model called DLA-MatchNet for remote sensing scenes from a perspective of discriminative representation and similarity metric. DLA-MatchNet consists of a feature learning module with an attention mechanism and an adaptive matcher. Cheng et al. [15] proposed a novel Siamese-prototype network (SPNet) to compute more accurate prototypes by elaborately designed self-calibration and inter-calibration, where label information of support samples and query samples are fully exploited. Cui et al. [16] designed a meta-kernel network (MKN) with a parameter linear classifier to learn the generalized features.

However, these existing few-shot remote sensing scene classification methods mainly focus on feature representations of query samples and the corresponding support samples based on the final output of the feature extractor. It can be regarded as one kind of metric learning, where in the embedding space similar classes should be close and dissimilar classes should be far apart. In this case, only high-level representations are taken into consideration and the inherent hierarchical features of convolutional neural networks (CNNs) are ignored which might be important to few-shot remote sensing scene

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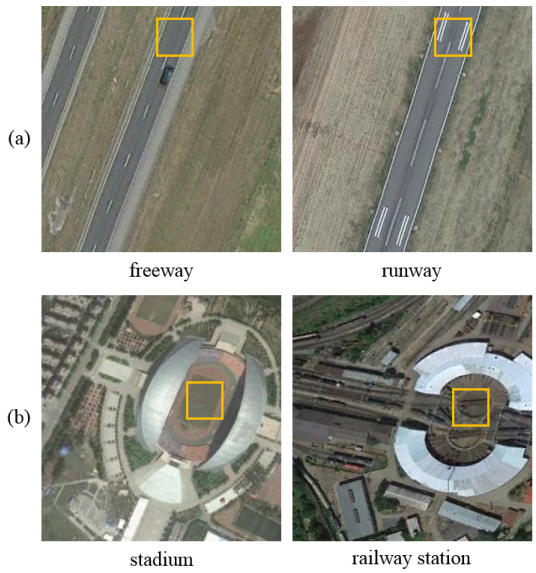


Fig. 1. Illustration of large interclass similarities. (a) Images of freeway and runway (b) images of stadium and railway station. The image pairs of freeway/runway and stadium/railway station contain some similar ground objects and look very indistinguishable from the overall perspective. However, there exist some dissimilarities on a smaller scale between these images, such as the "runway" image owning special runway lines and the "stadium" image having some green fields.

classification tasks. Let us take Fig. 1 for an example. The image pairs of freeway/runway and stadium/railway station contain some similar ground objects and look indistinguishable from the overall perspective. However, there exist some dissimilarities on a smaller scale between these images, such as the "runway" image owning special runway lines and the "stadium" image having some green fields. The learned features of the smaller scale would provide important clues for the following classifier to assign a correct label.

To address these problems, in this article, we propose a novel few-shot remote sensing scene classification method named hierarchical-relation network (HiReNet) to learn strong discriminative representations by leveraging the hierarchical features of a query sample and the corresponding support samples. The core parts of HiReNet are the proposed hierarchical relation learning (HRL) and feature aggregation with channel attention (FACA). Different from the usual methods, HRL can utilize hierarchical representations extracted by the embedding network so that more information can be incorporated to compute related similarities. In particular, we introduce transformer decoders in each relational extraction module (REM) to capture long-term mutual dependencies based on the inherent cross-attention structure. Moreover, the FACA module is carefully constructed to extract better representations, which consists of a local feature fusion and channel attention. Experimental results show that our proposed method can obtain better few-shot remote sensing classification outcomes than some state-of-the-art.

[*] todo here: the contributions should be revised later.

The main contributions of this article are summarized as follows.

1) We propose a new method named HiReNet for few-shot remote sensing scene classification to make full use of hierarchical features to learn a better metric representation. Our HiReNet can obtain state-of-the-art few-shot classification results on three public remote sensing datasets.
2) We introduce hierarchical-relation learning to leverage multilevel representations extracted by an embedding network. Meanwhile, we design a local feature fusion in basic modules (BMs) and channel attention is employed to make the fusion more discriminative. Ablation experiments verify the effectiveness of this design.

The rest parts of this article are organized as follows. We give a brief description of the background and related work of the few-shot remote sensing image scene classification in Section II. In Section III, we carefully describe our method and the proposed improvements. Many comparative experiments on two public remote sensing datasets and ablation studies are presented in Section IV. Finally, conclusions are drawn in Section V.

II. Background and Related Work

A. Deep Learning-Based Remote Sensing Image Scene Classification

In recent years, deep learning has made great progress in remote sensing image scene classification. Chaib et al. [17] introduce discriminant correlation analysis to perform convolutional feature fusion in order to enhance remote sensing scene classification accuracy. He et al. [18] proposed a new multilayer stacked covariance pooling (MSCP) model to automatically combine multilayer feature maps extracted by a pretrained CNN. Lu et al. [19] constructed a multisource compensation network (MSCN) to alleviate the challenges including distribution discrepancy and category incompleteness for remote sensing cross-domain scene classification. Cheng et al. [20] proposed discriminative CNNs from the perspective of metric learning to solve the problems of intraclass diversity and interclass similarity. Minetto et al. [21] introduced a new remote sensing classification framework named Hydra which is an ensemble of many CNN models. More recently, Chen et al. [22] devised a contextual information-preserved architecture learning (CIPAL) framework to utilize the contextual information in the remote sensing image scenes. Ma et al. [23] proposed a novel supervised progressive growing generative adversarial network (SPG-GAN) where the generative adversarial network is employed to generate the remote sensing image samples with label categories. Different from these methods, our work aims to enhance the feature representation of the few-shot learning backbone with feature aggregation and channel attention, and thus provide better hierarchic representations for the following relation learning part.

B. Few-Shot Learning

Few-shot learning aims to recognize unseen classes with very few labeled samples. In recent years, many impressive works have been proposed in this field. Snell et al. [24] proposed a prototypical network to compute prototype representation for few-shot classification. Sung et al. [25] designed a relation network to learn a deep nonlinear distance metric

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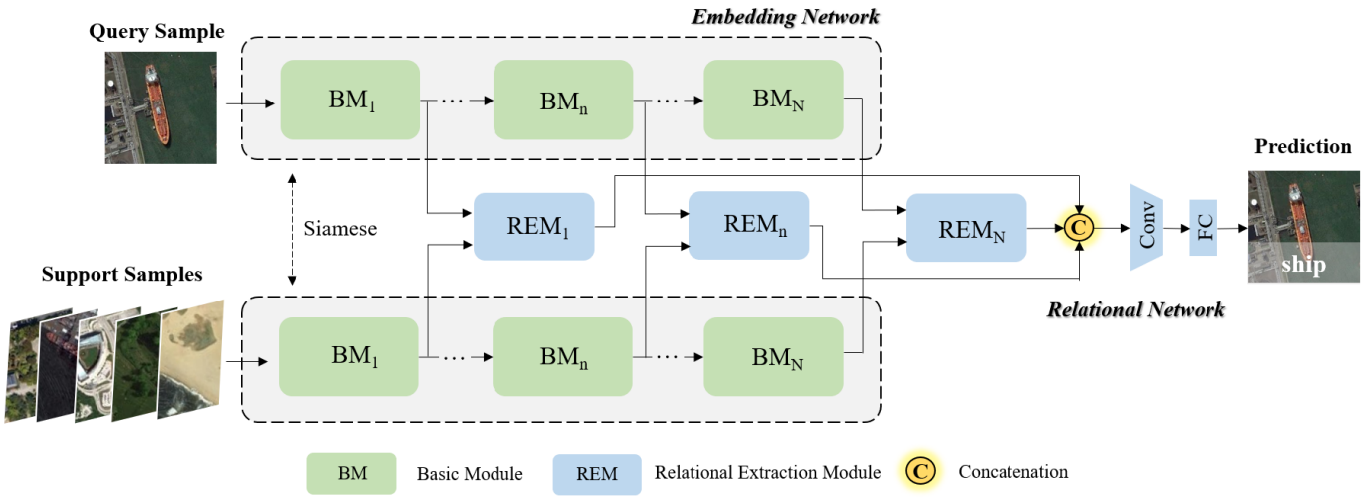


Fig. 2. Framework of the proposed method. Here we take a five-way one-shot scene classification as an example for illustration. Our method is combined with one embedding network for representation learning and one relation network for relation extraction between the query/support samples.

based on meta-learning. Chen et al. [26] explored the relationship between meta-learning and pretrained classification on the whole label-set, and introduced a simple meta-baseline to obtain competitive performance. Xie et al. [10] introduced the Brownian distance covariance and formulate it as a pooling layer into a deep network for the few-shot learning classification task.

More recently, few-shot learning has been introduced into the field of remote sensing. Gao et al. [27] proposed a deep few-shot relation network for the hyperspectral image classification with only a small number of labeled samples. Li et al. [28] combined few-shot learning with cross-domain to overcome domain shift for the hyperspectral image classification task. Besides, Li et al. [29] introduced meta-learning into the field of few-shot object detection based on the YOLOv3 architecture. Cheng et al. [30] proposed a remote sensing few-shot object detection method called prototype-CNN to leverage class-aware prototypes. Meanwhile, many researchers have developed few-shot learning into remote sensing scene classification [11], [14], [15], [16]. Different from these existing few-shot remote sensing scene classification methods, our proposed method aims to make full use of the inherent hierarchical features of the embedding network and learn better discriminative powers.

III. Methodology

A. Overview of the Proposed Method

The overview of the proposed method for few-shot scene classification is shown in Fig. 2. The HiReNet consists of an embedding network and a relational network, where the embedding network is designed to perform feature embedding of query/support samples and the relational network calculates the similarities between sample embeddings. We use many BMs to construct the embedding network to learn more discriminative representations. Following some few-shot learning methods [11], [14], [15], [16], the embedding network is utilized in a Siamese manner [31], and the query/support samples are extracted by the same model. Different from the usual methods, we introduce hierarchical relational learning to utilize hierarchical representations extracted by the embedding network so that more information can be incorporated to compute related similarities. The relational network is composed of many REMs where transformer decoders are employed to capture long-term mutual dependencies based on the inherent cross-attention structure. Meanwhile, we design an operation of FACA to extract stronger representations. More details about the embedding network and the relational network are carefully presented in the next two subsections. More details about FACA and hierarchical relational learning will be carefully provided in the next two subsections.

Given seen and novel remote sensing scene datasets D_{seen} and D_{novel}, HiReNet aims to learn feature representation and similarity relation based on D_{seen} and quickly classify the N scene classes with K labeled samples per class, which is called a N-way K-shot problem. In this article, D_{seen} contains training set D_{train} and val set D_{val}, and D_{novel} is equivalent to test set D_{test}. Importantly, these three sets are nonoverlapping with each other, i.e., $D_{\text{train}} \cap D_{\text{test}} = \emptyset$ and $D_{\text{val}} \cap D_{\text{test}} = \emptyset$.

B. Feature Aggregation With Channel Attention

As shown in Fig. 2, the embedding network consists of several BMs that perform deep embedding of the query/support sample. The detailed structure of BM is illustrated in Fig. 3. In order to extract more discriminative features, we propose a FACA structure in the BM where multilevel features are aggregated and channel attention is incorporated to re-weight these features.

At the head of BM, one convolutional layer with a pooling operation is implemented to reduce the spatial resolution of feature maps. Then several different level features are extracted and aggregated to obtain a more representative embedding

$$F_{\text{cat}} = \left[f_1, f_2, f_3\right]. \tag{1}$$

Following SENet [32], the aggregated features F_{cat} will be used to generate channel-wise statistic z via a global average pooling at a squeeze step. Specifically, let the spatial

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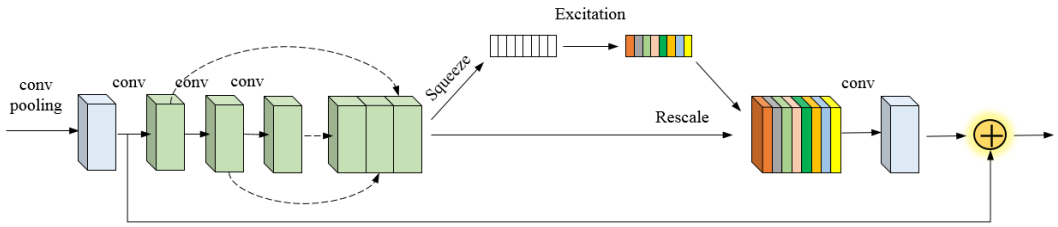


Fig. 3. Structure of the BM.

dimension of F_{cat} is $H \times W$, z is thus determined by the following equation:

$$z = H_{\text{AGP}}(F_{\text{cat}}) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} F_{\text{cat}}(i, j) \tag{2}$$

where $F_{\text{cat}}(i, j)$ denotes the activation value at the position (i, j), and H_{AGP} denotes the global average pooling operation. However, this channel descriptor does not reflect the relation of different channels. To capture the channel-wise dependencies, we utilize two fully connected layers to learn a sigmoid attention vector

$$s = \sigma(W_2(\delta(W_1 F_{\text{cat}}))) \tag{3}$$

where the δ is ReLU function and the σ is sigmoid function. Here W_1 can perform a channel shrink to control model parameters, and W_2 is used to increase channel dimensions.

The s reflects the nonlinear interaction relationship of the aggregated features of different levels. Then the final aggregated feature can be obtained

$$\hat{F}_{\text{cat}} = F_{\text{cat}} \cdot s. \tag{4}$$

Following the re-scaled features, one convolutional operation is employed to perform channel reduction with the same channel size as inputs. Moreover, it should be noted that we use a long skip connection after the pooling operator to ease network training and allow the FACA structure to focus on more informative parts.

C. Hierarchical Relation Learning

Apart from the embedding network, how to effectively obtain relation scores of query/support samples is very important for the few-shot classification problem. However, most existing few-shot remote sensing scene classification methods only utilize high-level embeddings of remote sensing images to learn a similarity relation and ignore the inherent hierarchical features which might be important to distinguish some scenes with large interclass similarities. To this end, we design a special relational network with HRL to fully leverage multilevel representations, which is the core part of our proposed method.

Specifically, the overall structure of the relational network is shown in Fig. 2. The relational network is composed of several REMs where each one is responsible for relation learning between query/support samples at one certain embedding stage. Suppose that E_Q^i denotes stage i feature embedding of the query sample and E_S^i denotes the corresponding embedding of the support sample. The REM_i will take these two embeddings as inputs and extract highly abstract relations.

The next question is how to design an effective structure of REM to leverage the embeddings of query and support samples. One common way is to utilize some convolutional layers to extract the relations of these two inputs. However, in this case, only local relationships are exploited and it is limited to express the similarity between query and support features. To address this problem, we elaborately designed the structure of the REM by combining it with transformers. Compared with traditional convolution, the transformer can effectively capture long-term dependencies and learn more discriminative representations [33].

The detailed structure of REM is illustrated in Fig. 4, we design a structure of two branches where each branch uses a transformer decoder with cross attention to process the query and support features, respectively. The learned representations of one certain stage would first be flattened and a linear projection is implemented to map the feature into a fixed dimension. Then the transformer decoders are to compute the mutual dependencies between the support and query representations. After the structure of the transformers, the learned features of query and support samples are concatenated together and two convolutional layers are used to learn a compact representation for the following relation computation. It should be noted that, in order to fully exploit the similarities, we design a mutual connection where the embeddings of query and support features are both the inputs of their transformer decoder.

The transformer decoders are the core components of the REM which is utilized to capture long-term dependencies of the query and support embeddings. Specifically, in the branch of query features, the query features extracted by the embedding network would be input into the following decoder, and at the same time, the support feature would also be introduced into this branch. The two branches of the REM are symmetrical and enable these two features to better capture similar and dissimilar information from each other. The multiheaded self-attention (MSA) module in the decoder is the same as cross-attention, where the attention between two inputs will be computed. Take q_0 and s_0 be the outputs of query and support features after the embedding network. The decoder output in the query branch can be obtained as follows:

$$\begin{aligned}
&q_0 = \left[f_{q_1}, f_{q_2}, \ldots, f_{q_N}\right] \\
&s_0 = \left[f_{s_1}, f_{s_2}, \ldots, f_{s_N}\right] \\
&q_i' = \text{MSA}(LN(q_{i-1})) + q_{i-1}, \quad i = 1, \ldots, L_d \\
&q_i'' = \text{MSA}\left(LN\left(q_{i-1}'\right), LN(s_0)\right) + q_{i-1}', \quad i = 1, \ldots, L_d \\
&q_i = \text{MLP}\left(LN\left(q_i''\right)\right) + q_i'', \quad i = 1, \ldots, L_d \\
&\left[f_{D_1}, f_{D_2}, \ldots, f_{D_N}\right] = y_{L_d}
\end{aligned} \tag{5}$$

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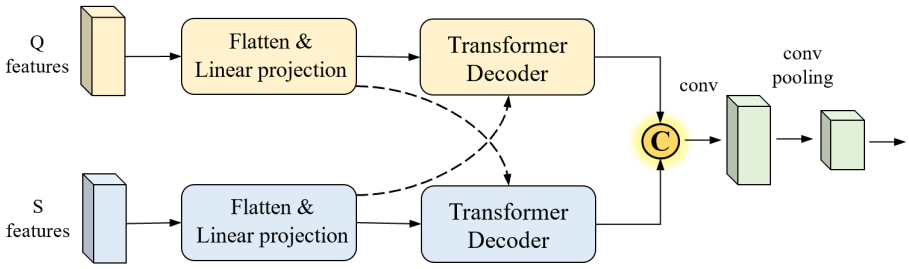


Fig. 4. Structure of the REM. Here we take REM_1 for an example.

where f_{D_i} is the decoder output, and L_d denotes the number of layers in the decoder. The MLP denotes multilayer perceptron and layer normalization (LR) [34] is used here before inputting features into MLP and MSA. Besides, the MLP has two layers with GELU nonlinear function [35]. In a similar way, the decoder output in the support branch also can be computed in the above way except the q_0 and s_0 are exchanged.

After relational extraction, the output of different REMs will be further combined as follows:

$$\text{RE}_{\text{cat}} = \left[\text{REM}_1\left(E_Q^1, E_S^1\right), \ldots, \text{REM}_i\left(E_Q^i, E_S^i\right), \ldots, \text{REM}_n\left(E_Q^n, E_S^n\right)\right]. \quad (6)$$

The concatenation of hierarchical relations will be used for the final computation of the similarity score via convolutional operation and some fully connected layers, where matched samples should get a similarity 1 and mismatched samples will have a similarity 0. The relation score r can be obtained as follows:

$$r = \sigma\left(\text{FC}(\text{Conv}(\text{RE}_{\text{cat}}))\right) \quad (7)$$

where the Conv represents the convolution operator followed by pooling and FC denotes the fully connected layer. The sigmoid function σ is used to constrict the relation score r into a reasonable range.

For the detailed structure of REM, we directly adopt several convolutional layers to extract local relations between the query/support samples in the embedding space. Convolution operation with a stride of 2 and pooling layer are both employed to control the size of relational features. This assures that the outputs of different REMs own the same spatial size for the following combination.

In this article, we regard the few-shot scene classification task as a regression problem and use mean square error (MSE) to train our model, where the matched query/support pair has the similarity 1 and the mismatched one has the similarity 0. The loss function can be defined as follows:

$$\text{Loss} = \sum_{i}^{N}\left(r_i - \mathbf{1}\left(\hat{y}_i = y_i\right)\right)^2 \quad (8)$$

where $\mathbf{1}(\bullet)$ is an indicator function. The label would be 1 when $\hat{y}_i = y_i$ and be 0 when $\hat{y}_i \neq y_i$. N denotes the total number of the meta-tasks.

D. Implementation Details

In our model, a convolutional block usually contains a convolutional layer with 3×3 kernel, a batch normalization, and ReLU function. The number of feature channels in the convolutional layer is set to 32. Meanwhile, a 2×2 pooling operation would be followed after the convolutional block when spatial reductions are needed. We set the number of BM to four and the total depth of the embedding network would be 20 convolutional layers. The number of REM thus is determined as four likewise.

In this article, following the most existing few-shot scene classification methods, we conduct five-way one-shot and five-way five-shot classification. Referring to [16], we train the model using the episodic training manner, where tens of thousands meta tasks are randomly sampled from the training set in total, and the input images are augmented by color jitter and random flip. For optimization, we use Adam [36] to train our model with the parameters of $\beta_1 = 0.9$, $\beta_2 = 0.99$ and $\epsilon = 10^{-8}$. The initial learning rate is set to 10^{-3}. At the test phase, 600 randomly generated episodes are used to compute the average accuracy on the test dataset. The HiReNet is implemented by Pytorch [37] and all the experiments are conducted on NVIDIA GeForce GTX graphics cards.

IV. Results

A. Dataset for Experiments

In order to verify the effectiveness of the proposed method, we use three typical remote sensing scene classification datasets including NWPU-RESISC45 [1], WHU-RS19 [38], UCMerced [39]. Following the setting of DLA-MatchNet [14], we split these datasets into a training set, a validation set, and a testing set shown in Table I to meet the requirement of few-shot learning.

1) NWPU-RESISC45 [1]. The dataset contains 45 different classes of remote sensing scenes including airplane, baseball diamond, beach, and chaparral with 700 images per class. The size of each image is 256×256. Following [14], the dataset is split into 25, 10, and 10 classes for training, validation, and testing which are carefully described in Table I.
2) AID [40]. The dataset consists of 10 000 images in total with 30 different scenes such as airport, bareland, baseball field, and beach. The image size is 600×600. We split AID into 15 training sets, five validation sets, and ten testing sets shown in Table I, also referring to [14].
3) UCMerced [39]. There are 21 classes in UCMerced dataset including agricultural, airplane, baseball-diamond, and beach. Each class owns 100 images with

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TABLE I
CLASSES SPLITED ON THREE REMOTE SENSING DATASET

Dataset	Training	Validation	Testing
NWPU-RESISC45	Airplane; Baseball diamond; Beach; Bridge; Chaparral; Church; Cloud; Desert; Freeway; Golf course; Harbor; Island; Lake; Meadow; Mobile home park; Mountain; Palace; Railway; Rectangular farmland; Roundabout; Sea ice; Ship; Sparse residential; Stadium; Wetland	Commercial area; Industrial area; Overpass; Railway station; Runway; Snowberg; Storage tank; Tennis court; Terrace; Thermal power station	Airport; Basketball court; Circular farmland; Dense residential; Forest; Ground track field; Intersection; Medium residential; Parking lot; River
AID	Airport; BareLand; BaseballField; Beach; Bridge; Commercial; DenseResidential; Desert; Farmland; Meadow; Pond; Port; RailwayStation; Resort; Stadium	MediumResidential; Mountain; Park; Parking; Playground	Center; Church; Forest; Industrial; River; School; SparseResidential; Square; StorageTanks; Viaduct
UC Merced	Agricultural; Baseball diamond; Buildings; Chaparral; Dense residential; Freeway; Harbor; Medium residential; Overpass; Parking lot	Airplane; Forest; Intersection; Runway; Storage tanks	Beach; Golf course; Mobile home park; River; Sparse residential; Tennis court

TABLE II
FEW-SHOT CLASSIFICATION ACCURACY ON THE NWPU-RESISC45 DATASET

Methods	5-way 1-shot	5-way 5-shot
Matching Net [41]	$37.28 \pm 0.99\%$	$52.31 \pm 0.73\%$
MAML [42]	$42.29 \pm 0.76\%$	$61.85 \pm 0.81\%$
Prototypical Network [24]	$43.99 \pm 0.78\%$	$69.35 \pm 0.66\%$
Meta-SGD [43]	$43.98 \pm 0.80\%$	$63.14 \pm 0.55\%$
LLSR [11]	42.20%	62.01%
Relation Network [25]	$44.55 \pm 0.79\%$	$68.21 \pm 1.39\%$
RS-MetaNet [44]	$52.78 \pm 0.09\%$	$71.49 \pm 0.81\%$
DLA-MatchNet [14]	$60.80 \pm 0.70\%$	$77.63 \pm 0.46\%$
MKN [16]	$65.84 \pm 0.89\%$	$82.67 \pm 0.55\%$
HiReNet(ours)	$70.43 \pm 0.90\%$	$81.24 \pm 0.58\%$

TABLE III
FEW-SHOT CLASSIFICATION ACCURACY ON THE AID DATASET

Methods	5-way 1-shot	5-way 5-shot
Matching Net [41]	$41.53 \pm 0.54\%$	$50.04 \pm 0.34\%$
MAML [42]	$47.93 \pm 0.72\%$	$61.79 \pm 0.75\%$
Prototypical Network [24]	$49.80 \pm 0.56\%$	$68.64 \pm 0.29\%$
Meta-SGD [43]	$49.97 \pm 1.23\%$	$64.84 \pm 0.97\%$
LLSR [11]	48.57%	62.15%
Relation Network [25]	$53.51 \pm 0.68\%$	$67.39 \pm 1.04\%$
RS-MetaNet	$56.32 \pm 0.55\%$	$74.48 \pm 1.11\%$
DLA-MatchNet [14]	$53.41 \pm 0.99\%$	$70.61 \pm 0.36\%$
MKN [16]	$57.29 \pm 0.59\%$	$75.42 \pm 0.31\%$
HiReNet(ours)	$59.43 \pm 0.66\%$	$74.12 \pm 0.43\%$

a size of 256 × 256 pixels. In our experiments, ten classes are used for training, six classes are selected for validation and the rest five classes are taken as testing following the setting of [14]. Table I lists the detailed information.

All the remote sensing images are resized into 256 × 256 pixels. The prevalent classification accuracy is considered as an evaluation metric for five-way one-shot and five-way five-shot, following some recent works [14], [15], [16].

B. Comparisons With Other Methods

We compare the proposed method on the NWPU-RESISC45, AID, and UCMerced datasets with some classical few-shot learning methods including Matching Net [41], MAML [42], Prototypical Network [24], Meta-SGD [43] and Relation Network [25], and with several state-of-the-arts of few-shot scene classification for remote sensing images such as LLSR [11], RS-MetaNet [44], DLA-MatchNet [14] and MKN [16]. It should be noted that here we take some classification results reported in the MKN [16] and use the same metric for a fair comparison. The overall results of the five-way classification tasks with one-shot and five-shot settings on NWPU-RESISC45, AID and UCMerced datasets are shown in Tables II–IV, where the accuracy is averaged over ten episodes with 600 randomly generated meta-tasks per episode from the test dataset.

Table II shows the few-shot classification accuracies of different methods on the NWPU-RESISC45 dataset. From Table II, it can be seen that our HiReNet can achieve the best performance for one-shot settings and the second best for five-shot settings. In the case of one-shot, HiReNet obtains 4.55% higher than the second MKN, 9.63% higher than DLA-MatchNet, and 17.65% higher than RS-MetaNet which they are specifically proposed for remote sensing images in recent years. The five-shot classification accuracy of our method is 1.43% lower than the best MKN, 3.61% higher than DLA-MatchNet, and 9.75% higher than RS-MetaNet.

The comparisons of one-shot and five-shot classification results of different methods on the AID dataset are listed in Table III. Our HiReNet achieves the best performance for one-shot settings and obtains comparative results for five-shot settings. For the one-shot, HiReNet obtains 2.14% higher than the second MKN, 6.02% higher than DLA-MatchNet, and 3.11% higher than RS-MetaNet. In terms of five-shot classification, the accuracy of our method is a little lower than MKN and RS-NetaNet and higher than other comparisons.

Moreover, we conduct some experiments on the UCMerced dataset to further verify the effectiveness of the proposed method. Compared to the two datasets mentioned above, the UCMerced is a small-scale dataset with fewer scenes and fewer images per category, posing a greater challenge for few-shot classification. As shown in Table IV, our HiReNet obtains the best performance for one-shot settings with 0.15% higher

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TABLE IV
FEW-SHOT CLASSIFICATION ACCURACY ON THE UCMERCED DATASET

Methods	5-way 1-shot	5-way 5-shot
Matching Net [41]	$42.22 \pm 1.02\%$	$52.32 \pm 0.92\%$
MAML [42]	$47.53 \pm 0.71\%$	$63.13 \pm 0.92\%$
Prototypical Network [24]	$50.23 \pm 0.61\%$	$70.54 \pm 0.20\%$
Meta-SGD [43]	$48.24 \pm 0.42\%$	$65.54 \pm 0.41\%$
LLSR [11]	47.95	63.32
Relation Network [25]	$52.43 \pm 0.87\%$	$68.65 \pm 0.95\%$
RS-MetaNet	$57.23 \pm 0.56\%$	$76.08 \pm 0.28\%$
DLA-MatchNet [14]	$53.76 \pm 0.62\%$	$63.01 \pm 0.51\%$
MKN [16]	$58.45 \pm 0.54\%$	$77.92 \pm 0.32\%$
HiReNet(ours)	$58.60 \pm 0.80\%$	$76.84 \pm 0.56\%$

than the second MKN, 4.84% higher than DLA-MatchNet, and 1.37% higher than RS-MetaNet. In the case of The five-shot classification, the accuracy of our method is 1.08% lower than the best MKN, 3.83% higher than DLA-MatchNet, and 0.76% higher than LLSR.

In summary, on these three datasets, our HiReNet achieves the highest accuracy in terms of one-shot and obtains comparative results in terms of five-shot. These results verify the effectiveness of the proposed method compared with the state-of-the-art. Meanwhile, it implies that HiReNet is especially suitable for the very few samples such as one-shot because HRL can fully leverage the hierarchical representation of the limited samples. More detailed discussions about the proposed improvements will be presented in Section IV-C.

C. Ablation Studies

In this section, we carry out some ablation studies on the NWPU-RESISC45 dataset to verify the effectiveness of the proposed components including FACA and HRL.

1) Compared With the Baseline: In order to visually observe the effectiveness of our method, as shown in Fig. 5, we plot the few-shot classification results of HiReNet and baseline with settings of five-way one-shot on the NWPU-RESISC45 validation set, where every 5k iteration there will be a validation. It should be noted that the baseline model consists of a normal embedding network and a relation network, where only the final embedding of query and support samples are used to determine the relation score. In Fig. 5, there is one validation data for every 5k iterations, and the solid line represents the results of our method, while the dashed line represents the baseline. It can be seen that compared with the baseline model, HiReNet can accelerate convergence speed, where the performance of the baseline at 40k iterations can be achieved after only 15k iterations by HiReNet. Meanwhile, after the entire training phase, our method finally achieves higher classification accuracy on the validation set compared to the baseline model. The validation curve of the proposed method is completely above the baseline curve, which fully proves the effectiveness of the proposed improvement.

2) Effectiveness of FACA and HRL: We further conduct experiments to demonstrate the effectiveness of the proposed FACA and HRL. We test the best checkpoint obtained by our methods with different settings and the baseline on the validation set, in order to compare their generalization ability on unseen classes of remote sensing images. The few-shot classification accuracies and 95% confidence intervals of five-way one-shot and five-way five-shot are reported in Table V. Compared with the baseline model, FACA and HRL components bring about significant improvements both in terms of one-shot and five-shot. Specifically, the FACA can obtain 1.2% higher in terms of one-shot and 2.18% higher in the case of five-shot than the baseline. The HRL further gains a 4.81% improvement for one-shot and a 2.79% improvement for five-shot. Apparently, HRL has a more significant performance improvement compared to FACA which will be carefully discussed in the following part.

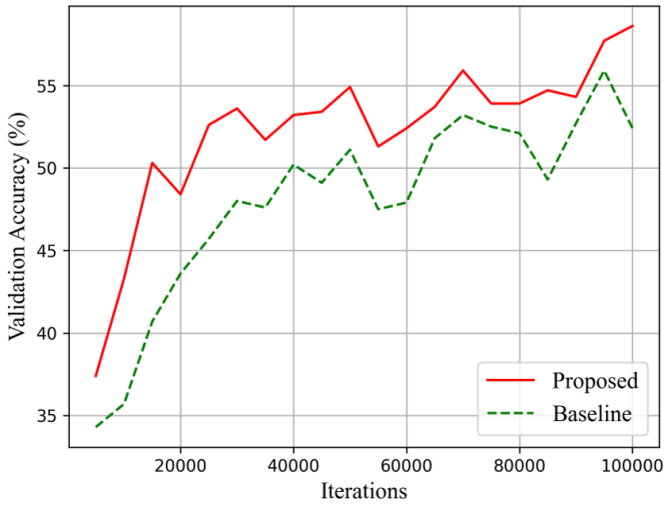


Fig. 5. Validation accuracy of the baseline and our method under five-way one-shot on the NWPU-RESISC45 validation set.

TABLE V
ABLATION STUDIES OF THE PROPOSED METHOD ON THE NWPU-RESISC45 DATASET

Methods	1-shot	5-shot
Baseline	64.42	76.27
Baseline + FACA	65.62	78.45
ours (Baseline + FACA + HRL)	70.43	81.24

3) Effects of Different Designs of HRL: We conduct experiments to explore the effects of different designs of HRL. Table VI lists the classification results of different structures of one-shot and five-shot on the NWPU-RESISC45 dataset. The number i indicates that the features of the embedding network at the ith stage will be utilized, and the first row is used as a benchmark for comparison, using only the highest-level embedding. An obvious phenomenon can be seen that the more levels of representations are used, the overall classification performance will be better. When all four levels are utilized simultaneously, our method can obtain the best few-shot classification results. Another interesting phenomenon is that in contrast to the five-shot, the structure of HRL brings a more significant performance improvement on the one-shot, which is consistent with the experimental results in the previous section. Compared to other methods,

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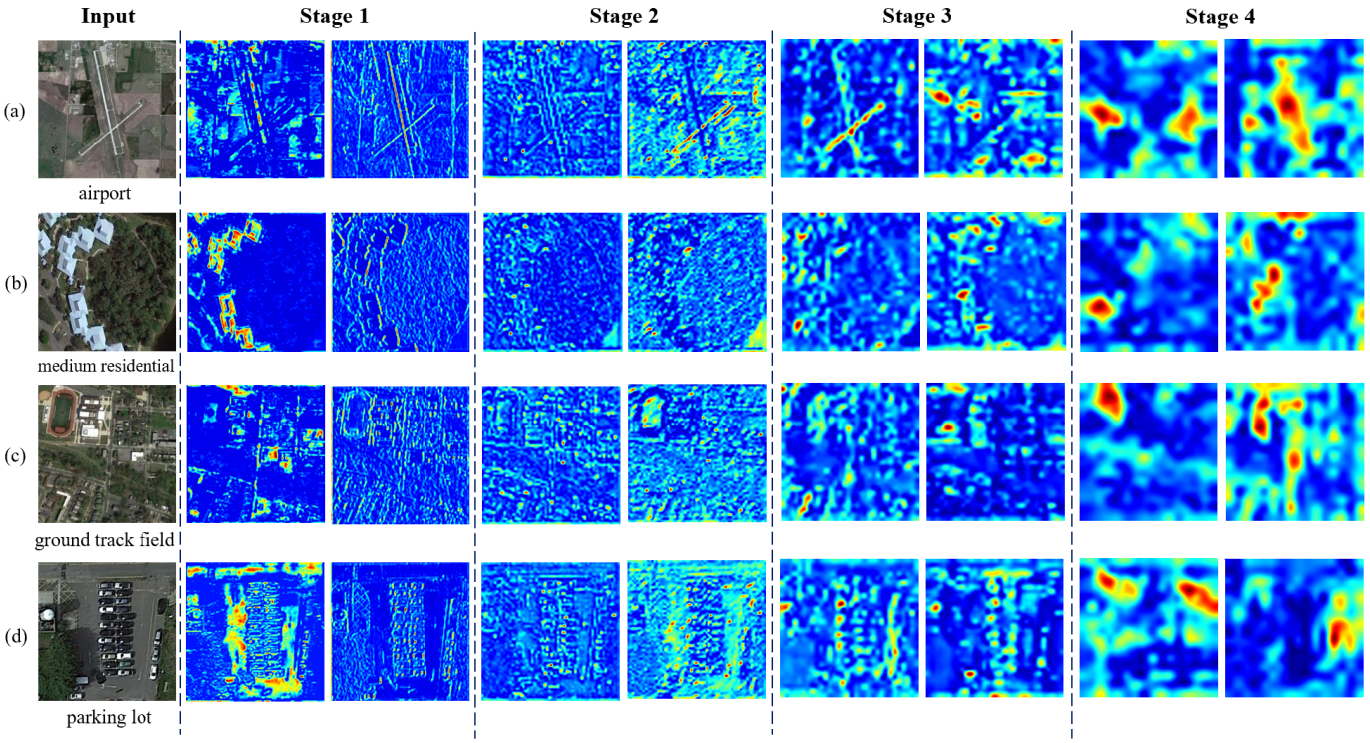


Fig. 6. Heat map visualization of features extracted by a trained embedding network given four images from different scenes as inputs. (a) Airport. (b) Medium residential. (c) Ground track field. (d) Parking lot.

TABLE VI
ABLATION STUDIES OF HRL STRUCTURE WITH DIFFERENT SETTINGS ON THE NWPU-RESISC45 DATASET

1	2	3	4	1-shot	5-shot
×	×	×	✓	65.62	78.45
✓	×	×	✓	68.82	80.70
✓	✓	×	✓	69.90	81.03
✓	✓	✓	✓	70.43	81.24

our method performs better when the number of samples is smaller. We speculate that when the number of samples is smaller, the contents in remote sensing images available are much more limited. By comprehensively utilizing information from different levels in the image, the proposed method can fully explore the comparative relationship between the query and support images, thereby enabling the relation network to obtain more accurate scores.

D. Network Visualization

The core of the proposed method is to leverage multilevel features extracted from the embedding network, where the feature at different levels focuses on different scales in remote sensing images. Through the design of HRL, comprehensive discrimination can be made by utilizing the features of different levels simultaneously. In order to better understand this point, we visualize the multilevel features extracted by the embedding network of the trained HiReNet, where some remote sensing images of different scenes are used.

We train a five-way one-shot classification model on the NWPU-RESISC45 dataset, and then randomly select four images from different scenarios in the test set, including airport, medium residential, ground track field, and parking lot. We input the corresponding images into the trained model and obtain feature maps of different levels from stages 1 to 4, with pixel sizes of 128 × 128, 64 × 64, 32 × 32, and 16 × 16, respectively. For better visualization, we randomly extract two feature maps from different stages and resize the feature maps to the same size as the input image of 256 × 256. The visualizations at different levels are shown in Fig. 6, where red indicates a higher response value for the corresponding position feature, and blue indicates a lower response value.

From Fig. 6, we can find that for the same scene, feature maps at different stages are diverse significantly. The receptive field of the higher stage is relatively large, and the response to some ground objects is obvious on the whole. At the lower stage, the feature map can better reflect textures, edges and distribution characteristics of the ground objects. The following phenomena can be observed.

1) Both high-level and low-level features can simultaneously contribute to the categorical information of ground objects. For example, as depicted in Fig. 6(a), the features extracted at stage 4 exhibit a distinct response around the airport runway, while at stage 1, the response along the airport runway line is more conspicuous. This suggests that for the airport, features at different levels can provide valuable information for few-shot classification.
2) High-level features can better represent the categorical information of ground objects. In the scene of a ground tracking field, as depicted in Fig. 6(c), the ground tracking field object is located in the upper left corner of the image, while the remainder contains many buildings and roads. The information of the target cannot be significantly reflected at the low-level feature map,

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as compared to the high-level feature map, where the response at the location of the ground tracking field is the highest. This indicates that high-level features can capture the ground object target in a complex background at a global level.

3) Lower-level features can better reflect the categorical information of ground objects. In the parking lot scenario at Fig. 6(d), global features are not easily learned, and in this case, lower stages can provide more edge and distribution-based features of vehicles, making it easier to recognize the features from the feature representation.

Based on these observations, we can conclude that the scales learned by different feature layers of the embedding network are variable, and significant features for different ground features may be present at the higher or lower levels. By comprehensively utilizing the features of different stages, the relation network can achieve better recognition of different types of remote sensing images. This further validates the effectiveness of the method proposed in this article.

V. Conclusion

In this article, we propose a novel few-shot remote sensing image scene classification method called HiReNet to learn stronger discriminative power by means of a designed HRL structure. The HRL aims to leverage the hierarchical representations of both query samples and corresponding support samples. Additionally, in order to extract effective embeddings, we introduce a feature aggregation with a channel attention structure to combine multilevel features and re-weight them. Ablation studies have verified the effectiveness of both FACA and HRL structures. Experimental results on three public remote sensing scene datasets show that our method can achieve better few-shot classification results in the one-shot setting than several state-of-the-art methods, and obtain competitive results in the five-shot setting.

Although HiReNet achieves strong overall performance, there are limitations to the proposed method. When simultaneously leveraging multilevel representations, more REMs will be constructed and involved. At the same time, the computation cost of relational network will significantly increase which is not conducive to practical applications. To this point, in the future, we will explore more efficient architectures that can adaptively utilize features from multilevels to reduce the complexity of the model.

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